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HISTORY OF EARLY ISOSEISMAL MAPS

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The first scientific tools of earthquake investigations were provided by isoseismic maps. The present paper describes the formation and development of cartographic representation of macroseismic information. Study of old isoseismal maps is of importance in assessing the earthquake hazard. Unfortunately there are only few well documented events prior the epoch of instrumental seismology for which the earthquake parameters (i.e. magnitude, focal depth) and tectonic position can be estimated. In the same time important steps forward in development of seismology (e.g. use of time observations, seismic speed determinations, use of geological information, birth of engineering seismology) are connected with attempts of representation of macroseismic data on maps.

Keywords: intensity scale; isoseismal map; isoseista

1. Introduction

It was recognized during the early stage of seismological investigations that for interpretation of the earthquake phenomenon the spatial distribution of its effects are of significant importance. Dolomieu (1784) and other investigators of the Calabrian earthquake (1783) found: the surface effects caused by a seismic event have a spatial distribution which depends on the types of rocks at the surface. In case of this earthquake in 1783 Schiantarelli (Musson 1994) made a quantification of damage: on his map he applied different symbols for settlements according to the amount of observed damage. The first earthquake map which consists isoseismal line was completed by P Kitaibel and A Tomcsányi, professors of Pest (today Budapest) University in 1814. Their two maps beside the isoseismal line circumscribing the

area of strongest damages which occurred during the Mór earthquake (14 January 1810, $I_0 \approx 8$) consists also fault lines generated by this event and the direction of the main shock.

In the paper by Egen (1828) the first attempt to complete a more sophisticated intensity scale is described on the basis of a moderate earthquake which occurred in Netherlands (today Belgium) in 1828. Egen's scale practically coincides with the first six lower degrees of the Rossi-Forel scale of 1883. Isolines were used on the maps of the Rhine basin seismic event (1846) (completed by Nöggerath in a co-operation with Schmidt) and of the Mid-German earthquake (1872) (completed by von Seebach).

Maps of Volger by Petermann (Volger 1856) and Mallet (1862) for the Visp valley earthquake (1855) and for the great Neapolitan event (1857) respectively comprise several lines of isoseists, but their intensity scales are still based on the data collected from a single seismic event alone.

The birth of isoseismal maps in their present-day sense is connected to the first intensity scales by Rossi (1874), Forel (1881), Forel-Rossi (1883) and Mercalli (1897) which are based on generalization of experience of many different seismic events. In present study the isoseismal maps are called "early" because they are based on not unified scales and they were completed in the pre-instrumental period of seismology, before the appearance of the uniformly used scales.

2. The first documented scientifically seismic event: the Calabrian earthquake (1783)

The great earthquake of Calabria (Fig. 1) which in 1783 was felt over two million square kilometres, killed 85 thousand persons and was followed by six disastrous aftershocks. This earthquake on the other hand was the first which was studied by scientific investigators. The first based on precise documentation cartographical representation were completed by Schiantarelli (Musson 1994) (Fig. 2), where different symbols were used for villages according to the amount of damage. Schiantarelli's map for a first time shows the seismic damage distribution.

Another very important scientific achievement is connected with the research activity after the tragedy of 1783: Dr. Domenico Pignataro completed one of the first earthquake catalogue of aftershocks which covered the time interval from 1783 to 1786. In Pignataro's list there are five intensity classes: slight, moderate, strong, very strong and violent. The number of shocks in this catalogue is 1186. The work of Pignataro comprises a part of investigation of Giovanni Vivenzio, royal physician of the royal court of Naples (Vivenzio 1786). Among the researchers of the earthquake of 1783 three names should be mentioned first of all: Grimaldi (Fig. 3), the secretary of war of the Neapolitan kingdom, Sir William Hamilton (Fig. 4) British envoy and volcanologist and Déodat de Dolomieu (Fig. 5), French geologist, who later on (in 1791) described the dolomite.

Grimaldi distinguished the five most violent events and gave a catalogue for Calabria from 1181 to 1756 (Delfico 1784). This catalogue was one of the first list of historical seismic events in modern sense. Sir W Hamilton was author of valuable



Fig. 1. Contemporary representation of 1783 Calabrian earthquake



Fig. 2. Detail of Schiantarelli's map of the the 1783 Calabrian earthquake (Musson 1994)

account on eruptions of Vesuvius. In 1783 he went to Calabria to observe the effects of the earthquake. He recognized the migration of the focus and concluded that the amount of damage depends on the geology and nature of the site. He published his memoir on the Calabrian earthquake in (Hamilton 1783). D de Dolomieu identified the dimension of the source, determined the displacement due to the earthquake



Fig. 3. D F A Grimaldi (1740–1784)



Fig. 4. Sir W Hamilton (1730–1803)



Fig. 5. D de Dolomieu (1750–1801)

and recognized the unequal distribution of the damages (1809). Although already after the great Lisbon earthquake of 1755 were constructed earthquake resistant constructions by Hungarian Carlos Mardel, the architect of the Portugal capital (who was working as a subordinate of famous marquis Pombal), the earthquake resistant buildings called “la casa baraccata” and introduced by the Neapolitan government after 1783 became effective and well-known tools to save human lives and properties for many centuries in Italy.

3. The isoseismal map of the Mór earthquake (Hungary, 1810)

The first map which consisted the isolines were completed by E Halley in 1702. This map was a geomagnetic map of the world. Even so the importance of isolines was not recognised in cartography during the 18th century. In the very beginning of the 19th century Alexander von Humboldt perceived first the importance of isolines and published his first geomagnetic intensity map (1804) and completed his meteorological map “Carte des lignes Isothermes” (1817), which was published however only in 1826. The first isoseismal map was published in 1814 by two professors of Pest (today Budapest) University, Pál Kitaibel and Ádám Tomcsányi, in their account on the Mór earthquake (Middle-Hungary), 14 January 1810 (epicentral intensity $I_0 \approx 8$). The Royal Governor Council ordered the University in Pest to investigate the effects caused by the earthquake. A Commission under the command of three professors (P Kitaibel – chemistry and mineralogy, Á Tomcsányi – physics and land-surveying, L Fabrici – agriculture and cadastre) visited the damaged territory. It was the first time, that a commission to investigate consequences of a seismic event was formed by earth scientists. The members of the commission steadily and consciously used in their work geological, hydrological and land-surveying methods of their epoch. The commission investigated the area between 3 and 17 February. Beside the inhabitant’s interrogatory it carried out hydrological observations, analysed the content of the spring-waters, measured the cracks generated in soil by earthquake and determined the geodetic position of observed anomalies. They finished their report on 17 March 1810. Due to the marked social interest the council of the University of Pest in 1814 decided to publish the book written by Kitaibel and Tomcsányi (Figs 6 and 7). As an attachment to the book (Fig. 8) entitled “Dissertatio de terrae motu Morensi anno 1810” (Kitaibel and Tomcsányi 1814) two carts with up-to-date for that time geographical background were prepared (Figs 9 and 10) which contained an isoline which surrounds the area with ruined or heavily damaged buildings.

The authors conclude their work with a quotation from Seneca: “There will be a time when our successors will be surprised that we had been unable to perceive a matter so easily understandable”.

4. Rhine-Netherlands, 23 February 1828: the first modern intensity scale of P N C Egen

A medium sized earthquake ($I_0 = 6$) occurred in Netherlands (the epicentral area of the event today belong to Belgium). This earthquake was described by P N C Egen (1793–1849) who was mathematician and physicist in Soest, Elberfeld und Berlin. He is author of many scientific contributions and books on mathematics, thermometers and electricity. His work on the Rhine-Netherlands earthquake (Egen 1828) (Fig. 11) consists two important contributions to seismology. Egen paper is accompanied by a map (Fig. 12), where the author underlined places with intensities VI and V with red colour, with blue and yellow colour those at which it was IV–III and II–I.



Fig. 6. P Kitaibel (1757–1817)



Fig. 7. A Tomcsányi (1755–1831)

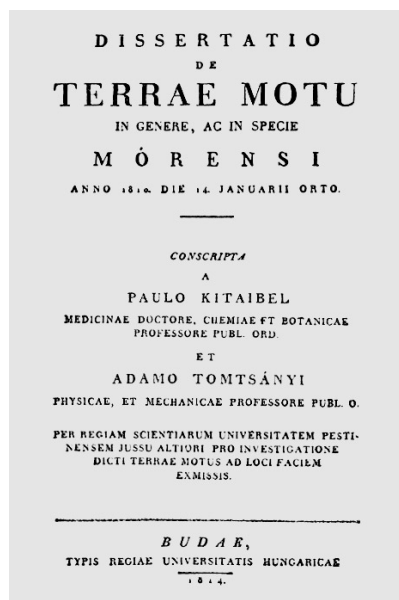


Fig. 8. The "Dissertatio de Terrae Motu..." (cover page)



Fig. 9. The first version of the isoseismal map of the Mór earthquake

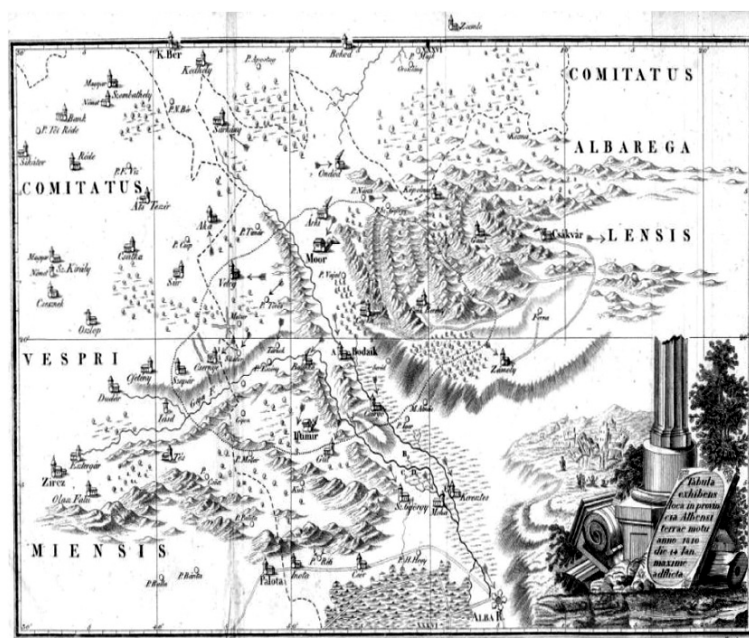


Fig. 10. The second version of the isoseismal map of the Mór earthquake

The first recognisable intensity scale in the modern sense of the word was drawn up by Egen in 1828; it was ahead of its time. It is very close to lower degrees of the De Rossi-Forel scale:

1. Only very slight traces of the earthquake are sensible.
2. A few persons, under favourable conditions, feel the shock; glasses close together, jingle; small plants in pots vibrate; hanging bells do not rung.
3. Windows rattle, house-bells rung; most persons feel the shock.
4. Slight movements of furniture; the shock in general so strong that it felt by everyone.
5. Furniture shaken strongly, walls are cracked, only a few chimneys fell down, the damage caused being insignificant.
6. Furniture shaken strongly; mirrors, glass and china vessels broken; chimneys fell down, walls cracked or overthrown.

5. Rhine basin earthquake, 29 July 1846. Results of Nöggerath and Schmidt

The central area of the earthquake of Rhine basin was close to Koblenz. It was felt in distance of some hundred kilometres from the central area of the quake (e.g. in Liège, Strasbourg, Stuttgart). The main investigator of this event was Nöggerath (Fig. 13) professor of mineralogy and mining at the University of Bonn. The main field of his activity lies in geology and mineralogy, but in the same time he published a series of contributions of seismological subject. The most important among them was the memoir on the earthquake in the region of Rheine in 1846 (Nöggerath 1847) (Fig. 14). The memoir is followed by a map with isoseismal lines (Fig. 15). The important achievement of Nöggerath's investigation was the first determination of the position of epicentre at St. Goar (close to Koblenz) with the use of the isoseismal lines.

The scale which can be derived from the map consists two scales:

1. District in which the shock was felt with the greatest intensity (continuous small circle of radius 45 km)
2. Places where the shock was felt (polygonal broken line and an outer circle, which surrounds these places).

The second important section of the monograph is the determination of velocities. This section was completed by Schmidt (Fig. 16) at that time assistant of the Astronomical Observatory of the University of Bonn. Later on (in 1858) after his appointment as director of observatory Athens he wrote besides his numerous contributions in astronomy an account of the earthquake in Zsolna (today Zilina in Slovakia) and his famous "Studien über Erdbeben" in which he introduced the term epicentre and main shock, furthermore completed a catalogue of earthquakes

of the south-eastern of Europe. In the monograph of 1847 Schmidt describes how he on the basis of his accurate time observations (Fig. 17) determined the speed of propagation of the earthquake effect.

Taking St. Goar as the centre, Schmidt found the main velocity ~ 500 m/s. In his study he used the modern mathematical tool of those days: the least squares method.

153

**XIII. Ueber das Erdbeben in den Rhein- und
Niederlanden vom 23. Februar 1828;
von P. N. C. Egen.
(Hiezu die Charte Taf. III.)**

Eine kleine Reise, die ich gegen Ende Februars mit Hrn. Prof. Hoffmann aus Berlin zu machen die Freude hatte, führte mich an die östlichen Gränzpunkte, welche das Erdbeben kurz vorher noch erreicht hatte. Diefes gab die erste Veranlassung, dafs ich mich um weitere Nachrichten über diefs in unseren Gegenden seltnere Naturereignifs angelegentlicher bemühte.

Es ist mir gelungen, über die Verbreitung des Erdbebens in seinen Hauptrichtungen ziemlich vollständige Nachrichten einzuziehen. Theils wurden mir diese durch die rheinischen und niederländischen Zeitungen, theils durch Privat-Correspondenz und mündliche Mittheilungen geliefert. Ich habe nur solche beachtet, die ich für zuverlässig hielt.

Um die Uebersicht zu erleichtern, habe ich auf einer kleinen Charte die Hauptdata des Erdbebens durch Zeichen darzustellen gesucht. Von allen auf der Charte genannten Oertern liegen mir Nachrichten, dafs dort das Erdbeben verspürt worden sey, vor.

Das Erdbeben trat kurz nach $8\frac{1}{2}$ Uhr Morgens ein. Zwei bis 4 Erschütterungen folgten unmittelbar auf einander, so dafs die Gesamt-Dauer nur wenige Secunden betrug. Nach einem Berichte aus Loewen soll dort die Dauer eine Minute gewesen seyn; in Tirlemont sollen die Stöße sogar 7 Minuten lang angehalten haben; in St. Tron legt man ihnen eine Dauer von 20 Secunden bei. Obgleich es nun möglich ist, dafs an diesen Orten, die im Centrum der Erschütterungen liegen, ihre Dauer länger war, als andwärts, so möchte doch die Furcht auf obige Angaben den bedeutendem Einfluß ge-

Fig. 11. First page of Egen's paper

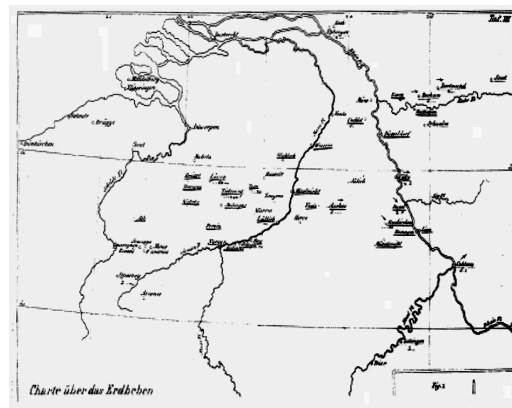


Fig. 12. The black and white copy of Egen's (1828) map. The earthquake was felt in settlements marked on the map



Fig. 13. J J Nöggerath (1788-1877)



Fig. 14. Cover page of Nöggerath's memoir

Nöggerath estimation of the distribution speed of Lisbon earthquake (1755) using J Michell's data was similar (~ 500 m/s).

6. Visp valley earthquake (25 July 1855), activity of Volger and Petermann

G H O Volger (1822–1897), professor of geology in Göttingen, Zürich und Frankfurt am Main was an acknowledged person both of political and scientific life of Germany. Despite of this situation he became admitted as a historian of Swiss earthquakes. He was a leading person of geological school of Neptunists. Due to this circumstance he had quite unrealistic opinion on the origin of the earthquakes, too. This did not prohibited him to complete an excellent account on the Visp valley earthquake (25 July 1855) (Volger 1856) (Fig. 18). To this essay A H Petermann

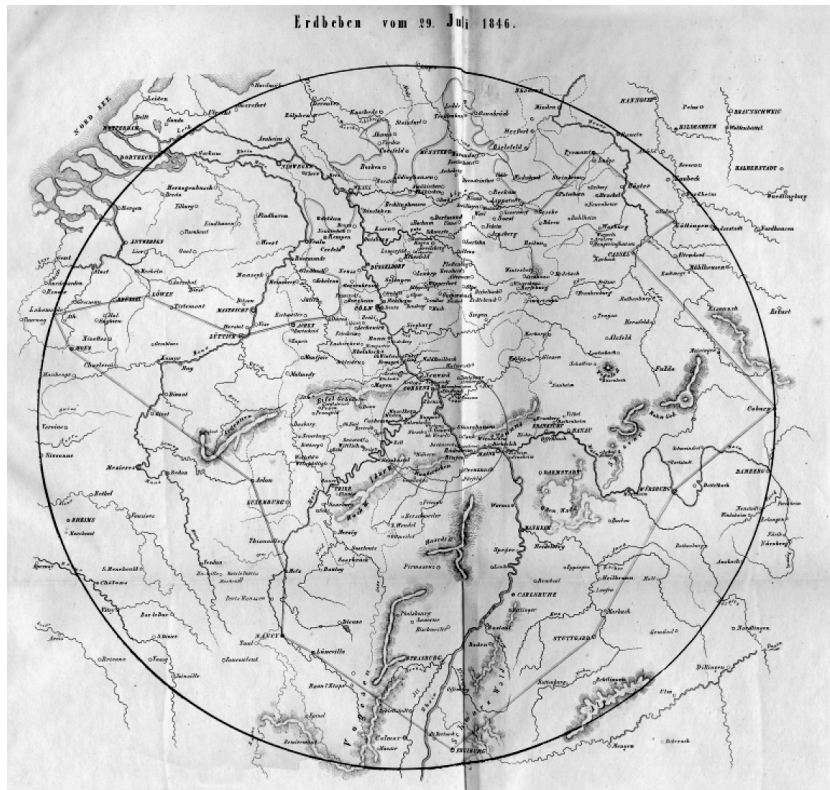


Fig. 15. Isoseismal map of the Earthquake in Rhein area by Nöggerath



Fig. 16. J F J Schmidt (1825–1884)

have prepared a modern isoseismal map (Fig. 19). The 1855 earthquake at Visp was the strongest in the last 300 years in Switzerland. It caused heavy damages in the region of the middle Valais. Volger collected together a substantial description of the damages. In such a way he was able to describe first the macroseismic scale on the basis of building damages. This way Volger founded the modern macroseismic investigations.

Bonn	t = 9 ^h 24'	mittl. Zeit.	p = 4	d = 10,12 pr. Meil.	$\mu = -2',50$
Aachen	23	" "	3	17,75	" " " - 6,58
Düsseldorf . .	26	" "	2	17,90	" " " - 3,78
Neuss	26	" "	1	18,00	" " " - 4,11
Elberfeld . . .	26	" "	2	17,10	" " " - 2,38
Coblenz	25	" "	2	3,25	" " " - 0,50
Ehrenbreitstein .	25	" "	1	3,15	" " " - 0,48
Boppard	25	" "	1	1,60	" " " - 0,50
Mainz	28	" "	2	5,75	" " " + 2,20
Darmstadt . . .	30	" "	2	9,80	" " " + 3,75
Karlsruhe . . .	31	" "	3	17,87	" " " + 2,72
Cronberg	31	" "	1	7,40	" " " + 3,08
Frankfurt . . .	32	" "	3	9,25	" " " + 3,88
Marburg	33	" "	3	13,87	" " " + 4,17
Pyrmont	9 ^h 39'	" "	2	30,25	" " " + 6,11

Fig. 17. Time observations of Schmidt and time differences relative to St. Goar

UNTERSUCHUNGEN ÜBER DAS LETZTJÄHRIGE ERDBEBEN IN CENTRAL-EUROPA.

Von G. H. Volger in Zürich.

(Nicht Tafeln 4 und 5.)

Das Studium mancher Phänomene der Physik unserer Erde hängt leider sehr wenig von der Neigung und Prädisposition der Forscher ab, indem die Gelegenheit zu dem wichtigsten Erfordernisse, zur Beobachtung, sich Manchem, der sie erschat, nie in genügender Weise darbietet, einen Andern dagegen gerade dann überrascht, wenn gleichsam alle Verhältnisse sich vereinigen, um ihm die Benutzung der günstigsten Gelegenheit unmöglich zu machen. Zu diesen Phänomenen gehören die meisten meteorologischen Vorgänge, deren Eintritt von dem Zusammenwirken so vieler verschiedener Bedingungen abhängt, dass dem mit diesen Bedingungen nicht Vertrauten der Kausalzusammenhang derselben völlig entgeht, wie bei jenen psychologischen Vorgängen, welche wir bildlich mit denen der Witterung zu vergleichen lieben, indem wir von der Launenhaftigkeit des Wetters und von der Wetterwendigkeit der Launen reden. Ebenso verhält es sich mit dem Phänomene der Erdbeben, welches übrigens in Wirklichkeit der Meteorologie vielleicht weit näher steht, als man bisher geneigt war, dieses zuzugestehen. Als daher im Juli des vorigen Jahres inmitten der Schweiz eins der heftigsten Erdbeben eintrat, von welchem wir Kunde besitzen, glaubte ich die Gelegenheit nicht unbenutzt lassen zu sollen, die Beobachtungen über dasselbe möglichst vollständig zu sammeln, um durch deren allseitige Prüfung zur Sicherung eines minder schwankenden Bodens, als man bisher ihn besaß, für die wissenschaftliche Anschauung der so räthselhaften Schwankungen des Verhältnisses aller unserer Begriffe von Festigkeit, des Erdbodens selber, beizutragen. Die Aufgabe, welche ich mir gesetzt hatte, wurde freilich unter meinen Händen. Die Ausdehnung des durch die heftigsten Stöße erschütterten Gebietes erwies sich bald durch die einlaufenden Nachrichten als eine sehr beträchtliche, weit über den Umfang der Schweiz hinausgehende, und selbstdemnach, was für die Ausbeute an Beobachtungen sehr wichtig war, gänzlich in den Bereich civilisirter Staaten, innerhalb deren nicht leicht eine irgend bedeutende Wahrnehmung verloren gehen konnte. Im Kanton Wallis aber, wo die heftigsten Wirkungen der ersten Stöße im Rhodan-Piemont's Geogr. Mittheilungen. 1856, Nr. III.

Thale von Veigy bis Visp und im Thale der Gornier-Visp von Visp bis St. Niklaus sich geäußert hatten, wiederholten sich die Erschütterungen sehr zahlreich mit kürzeren und längeren Pausen bis zum Ende des Jahres, so dass die Einzelheiten der gemachten Beobachtungen über die ganze Erschütterungsreihe und ihre Wirkungen ausserordentlich ansehnlich wurden. Dazu zeigte sich bald, dass, um eine feste Grundlage für die Schlüsse zu erlangen, mit deren Hilfe wir die Gesetze der Erdbeben zu verstehen und das Phänomen überhaupt als Funktion in der Physiologie des tellurischen Organismus zu erkennen suchen, die Sammlung von Beobachtungen und die übersichtliche Zusammenstellung solcher bei Ereignissen der Gegendart allein nicht genügt. Auch die Beobachtungen aus früheren Zeiten, so fragmentarisch dieselben immer sein mögen, ja, so sehr obendrein der Werth eines Theiles derselben durch die Befangenheit der Beobachter selbst vermindert wird, bieten eine reiche Spende von Belehrung dar und sind sogar das einzige Mittel, um manche Seiten dieses Phänomens zum Vorschein zu bringen. Kurz, wir müssen das Phänomen durch Raum und Zeit auf das Specieille verfolgen, um zu wissenschaftlichen Resultaten über dasselbe zu gelangen. Aber nicht die Anhäufung von Thaten allein genügt; ebenso wichtig, ja fast noch wichtiger als sie, ist die Sondernng derselben, ebenfalls nach Raum und Zeit, die Zusammenfassung der Resultate in gewisse nach diesen Grundverhältnissen geordnete Gruppen.

Man hat bis jetzt, wenn es sich nicht um eine offenebare vulkanische Beschaffenheit des Bodens handelte, die geognostische Untersuchung der von einem Erdbeben betroffenen Gegend vernachlässigen zu dürfen geglaubt, indem man durch Vorurtheil annahm, dass die Ursachen der Erdbeben sich in Tiefen des Erdbodens befinden, welche unseren Untersuchungen unzugänglich seien. Allein die Häufigkeit der Erdbeben in gewissen, zum Theil sehr eng umgrenzten Gegenden liess vermuthen, dass lokale Verhältnisse dieselben, wenn nicht bedingen, so doch begünstigen. Dabei zeigte sich denn bald, dass das relative Alter der in solchen Gegenden herrschenden Formationen, mit dessen

Fig. 18. First page of Volger's contribution on the Visp valley earthquake

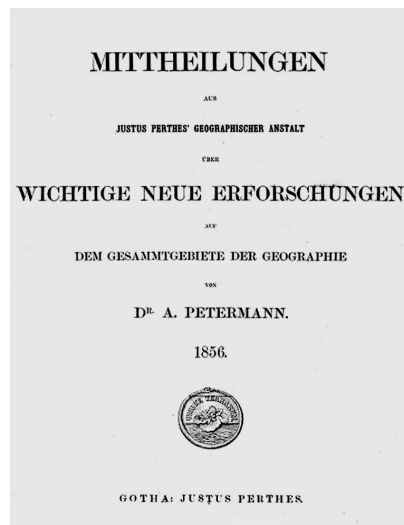


Fig. 19. Title-page of the "Mitteilungen" (editor A H Petermann)

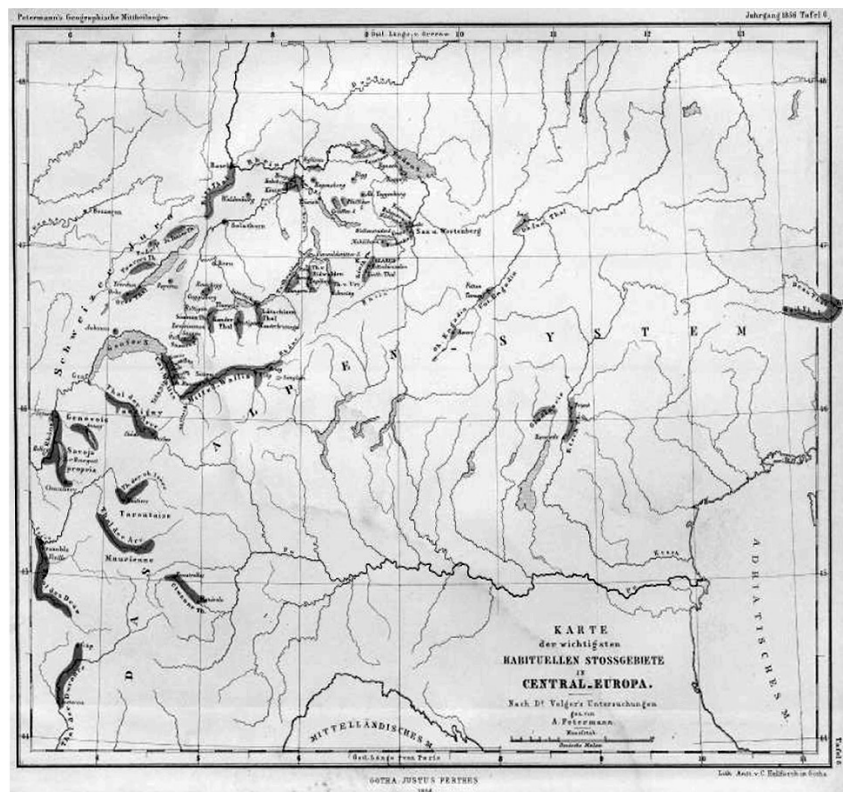


Fig. 20. Volger's seismoactive zones of the Alps

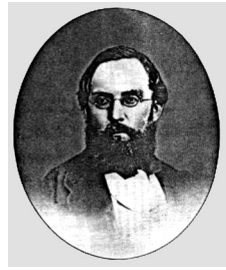


Fig. 21. A H Petermann (1822–1878)

For the first time the description of a seismic event was based on geological and tectonic background. Volger gives the description of seismo-active zones in the Alps (29 zones are mentioned by him) (Fig. 20). On the basis of data provided by Volger Petermann (Fig. 21) completed an isoseismal map (Fig. 22), the first where the isoseismal lines depend on well-documented building damages:

1. Whole houses and churches fell.
2. The strongest parts of buildings damaged.
3. Slight damages to buildings.
4. Small damages of buildings.
5. The shock certainly felt.

Beside the monograph on the Visp valley earthquake Volger's most important contribution to seismology was the Swiss earthquake chronicle consisting more than 1300 seismic events (Volger 1857: "Untersuchungen über das Phänomen der Erdbeben in der Schweiz", 1855).

From many points of view similar to Volger's work was carried out in Hungary by L H Jeitteles (1830–1882), Austrian zoologist, who was well known for his dispute with Charles Darwin. Beside zoology he published papers on very different subjects (e.g. one of his contributions treats musical matter: "Petőfi und die ungarischen Volkslieder, Stimmen der Zeitbahn", 1861). Among his accounts some discuss seismological subjects written mainly in German and also in Hungarian. During his stay in Kassa (today Kosice in Slovakia) between 1858–1861 he published a paper on seismicity of Hungary (in Hungarian) and a book on the Sillein (Zsolna in Hungarian, Zilina in Slovakian) earthquake (15 January 1858). In his discourse Jeitteles (1859) shows the influence of topography and geology on the intensity of the shock. For example he found that the intensity increased on not so dense more recent rocks, and was lower on the crystalline rock formations. Jeitteles map of the earthquake contains three isoseismal lines which encircle the areas of strong, distinct and weak seismic activity. In this work was introduced by Jeitteles the term "seismological centrum" identical to presently used epicentrum.

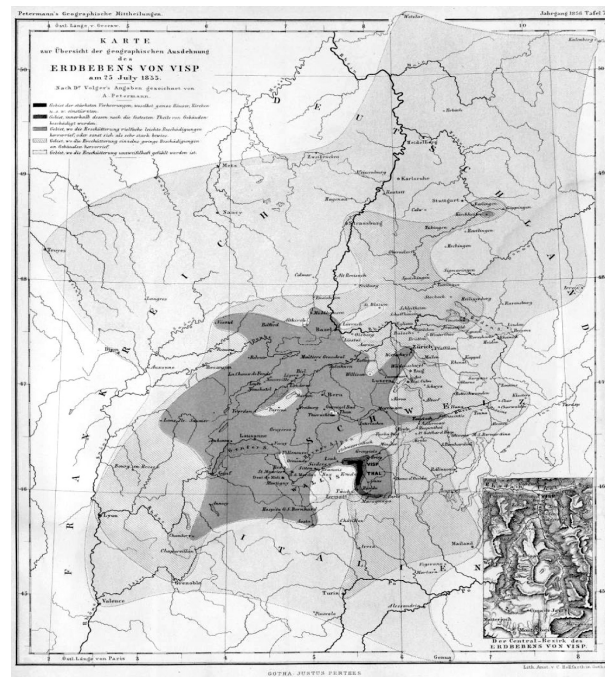


Fig. 22. Petermann's map of isoseismal lines for the Visp valley earthquake (22 July, 1855)

7. Neapolitan (Padula) earthquake (16 December 1857) and Mallet's "The first principles of observational seismology" (1862)

The area around Padula, southern Italy was devastated by a very strong earthquake in which 11,000 people were killed (Fig. 23). At that time this earthquake was the 3rd largest known quake in the world and was reckoned to be 7 on the Richter scale. The world-famous investigator of the earthquake was R Mallet from Ireland (Fig. 24). He collected the most precious data for determining the velocities, shock directions and depth of the origin of the shocks. His monumental treatise (Fig. 25) consists a lot of new ideas, is well illustrated and affixed with a wealth of maps and plates.

His map of isoseists (Fig. 26) consists three lines. Consequently the scale of Mallet is formed by four degrees:

1. Most of the towns are prostrated.
2. Large parts of buildings were thrown down and persons were killed.
3. Slight damage to buildings occurred without any loss of life.
4. The shock was perceived.

The remarkable features of the seismic map completed by Mallet are:

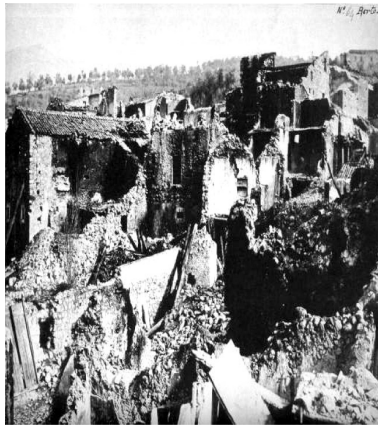


Fig. 23. Padula, 1857

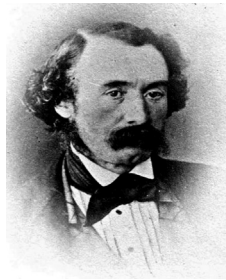


Fig. 24. R Mallet (1810–1885)

- On the basis of building damages and clefts the position, the depths and the dimension of the source were obtained. The lines in the middle of the map show the lines of the wave-path.
- Remarkable is the eccentric position of the epicentre.

8. Mid-German earthquake, 6 March 1872, the Seebach's monograph

The main and single contribution of von Seebach (1839–1880) (Fig. 27) is given in his monograph “Das mitteldeutsche Erdbeben vom März 1872, Ein Beitrag zu der Lehre von den Erdbeben”, Leipzig, 1873 (Fig. 28).

Seebach introduced in seismology the accurate time-observations. He drew two maps. The first shows the pleistoseist zone and the isoseists (Fig. 29) and the other the homoseists (Fig. 30).

The three isoseist lines are:

- area within which building damages occurred,
- accompanying sounds were observed,
- limits of the disturbed area.

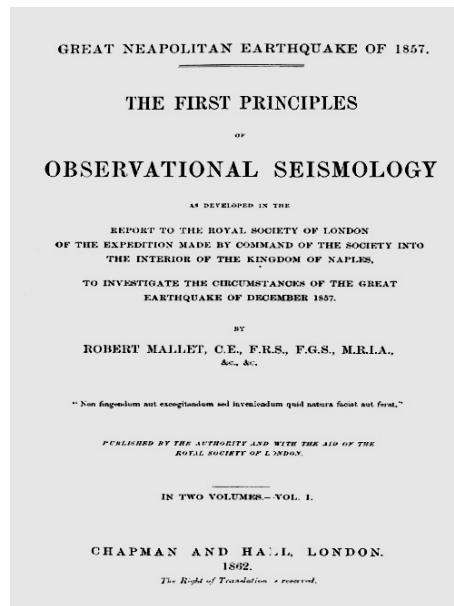


Fig. 25. Mallet's treatise

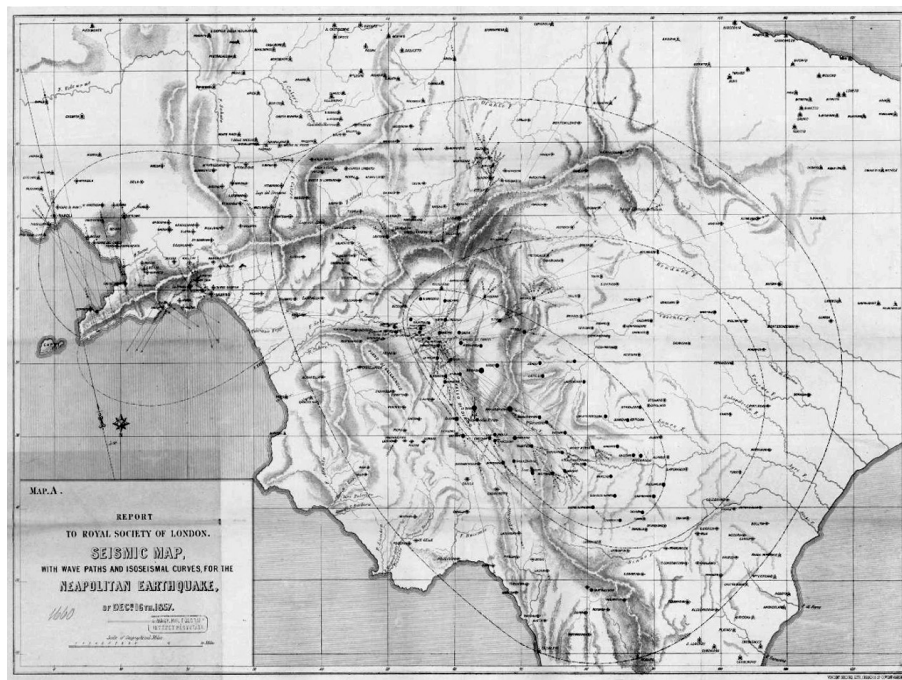


Fig. 26. Seismic map with wave paths and isoseismal curves for the Neapolitan earthquake of December 16, 1857 (Mallet 1862)

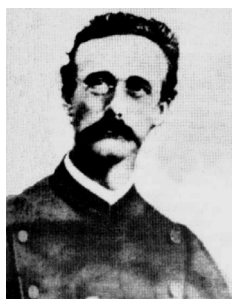


Fig. 27. K A L von Seebach (1839–1880)

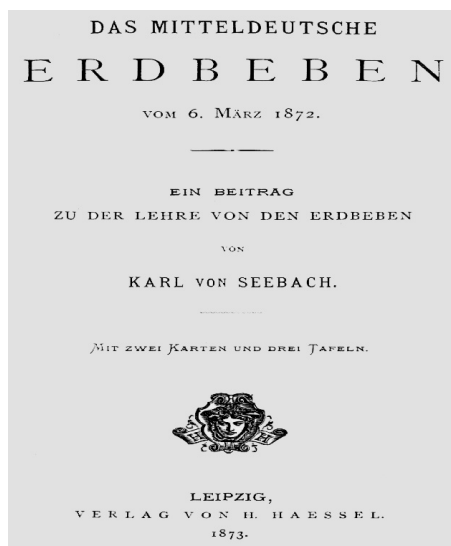


Fig. 28. Front-page of Seebach's monograph

Seebach's main contribution to seismology beside the accurate time-observations is:

- determination of epicentre with the use of homoseists,
- method of determining the depth of the focus,
- the first hodograph curve,
- calculation of mean seismic wave velocity (~ 800 m/s).

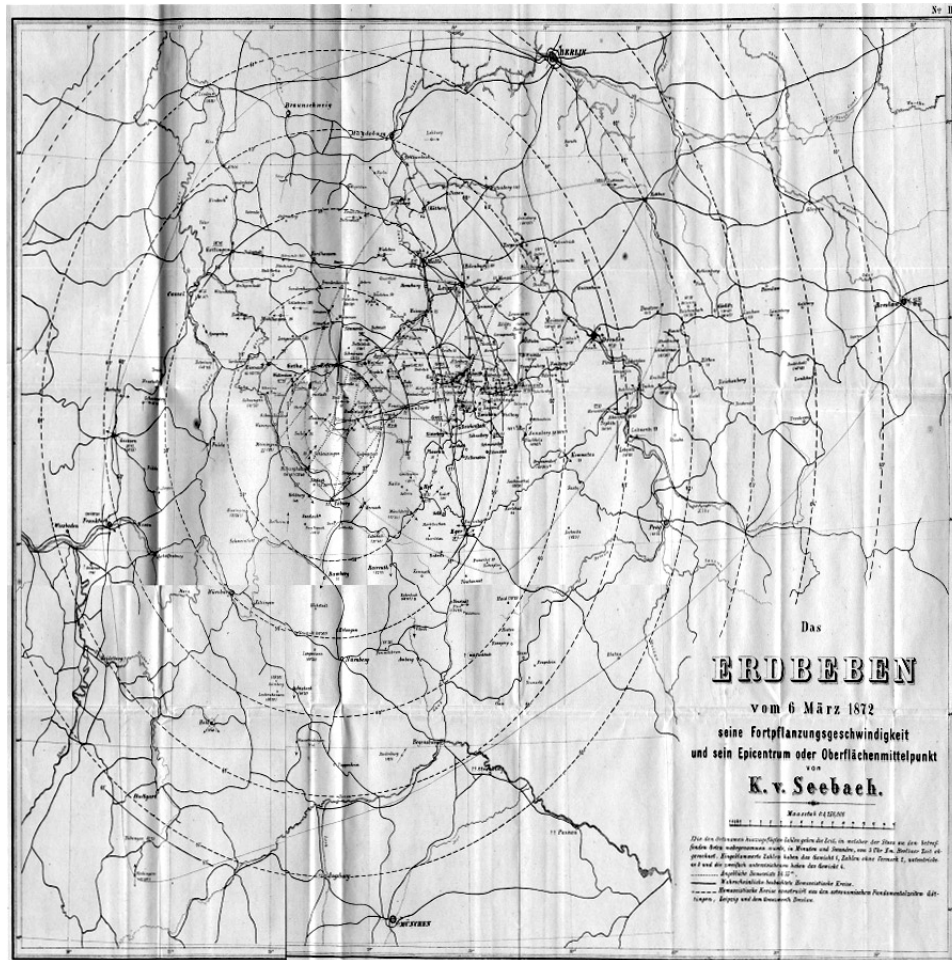


Fig. 30. Map of homoseists (the homoseist line connects places, where the earthquake arrives in the same time (Sieberg 1923))

The most important input of early isoseismal maps is that they used first regularly the isolines. The isolines are of great importance in natural sciences, probably first of all in earth sciences. It was shown above, that they were used consciously first in seismology. Additionally the authors of early isoseismal maps introduced in seismology the accurate time measurement and they determined the speed of propagation of seismic effect. On the basis of macroseismic data they assay the focal depth of the earthquake foci. Already the early workers of seismology in 18th century perceived the importance of geological, hydrological agents in enhancement of seismic hazard. After the Calabrian earthquake de Dolomieu and Hamilton carried out first geological observations in seismology. In 1810 the Royal University of Pest sent already an expert commission of earth scientists to region of Mór earthquake, who



Fig. 31. Hand made isoseismal map of Jászberény earthquake. In zone I the strongest building damages occurred, in zone II slight damages took place, in zone III the earthquake was felt. Arrows show the shock directions

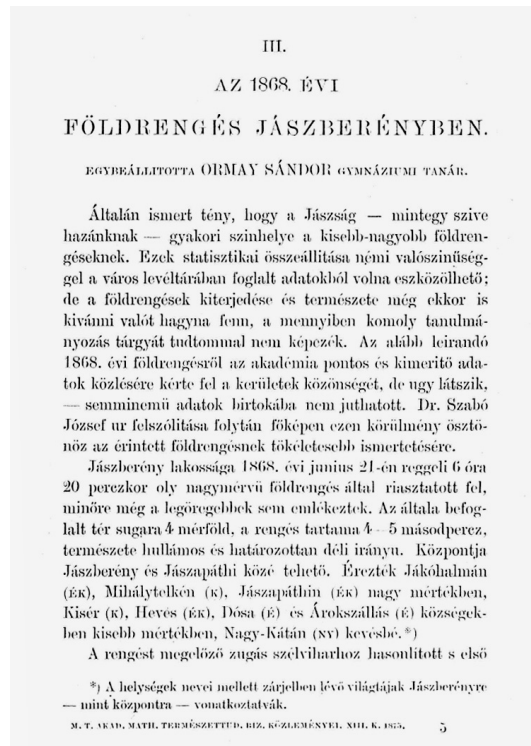


Fig. 32. The first page of Ormay's (1875) work on the Jászberény event

made detailed geological observations, geometrical (geodetic) measurements, carried out chemical analysis of spring waters and observed their output. In 19th century the study of the earthquake phenomenon consciously was connected to geological structures. The development of seismological terminology was in marked extent connected with time of early isoseismal maps. For example terms “main shock” and “epicentrum” can be found first in contributions written by J F J Schmidt, R Mallet introduced the words “isoseismal lines”, “angle of emergence” and “origin of disturbance”.

Consequently the contribution of early seismologists to development of earth sciences and particularly to advancement of seismology is very important.

Additionally the importance of early isoseismal maps is connected with the fact that on the basis of some well-documented early studies it is possible to estimate the magnitudes, damage fields and focal mechanisms of pre-instrumental seismology.

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References

- Baratta M 1910: La Catastrofe Calabro-Messinese, 28 dicembre 1908, 214–215.
 Bulletino del Vulcanismo Italiano, 10, 1883, 67–6–8; Archives des Sciences Physiques et Naturelle, 11, 148–149, 1884 (Rossi-Forel scale)
 Davison C 1921: *Bull. Seism. Soc. Am.*, 11, 93–129.
 Davison C 1927: Scales of seismic intensity: supplementary paper, 16, 4, 158–166.
 De Rossi 1878: Bulletino del Vulcanismo Italiano, 46.
 Delfico M 1784: “Eulogio del Marchese D F A Grimaldi”, Naples
 Dolomieu D 1784: Mémoire sur les tremblements de terre de la Calabre pendant l’année 1783, Rome
 Dolomieu D 1809: A dissertation on the earthquakes in Calabria Ultra in the year 1783 by the commander D de Dolomieu (Translation in Pinkertons Voyages and travels), Vol. 5, 273–279.
 Egen P N C 1828: Über das Erdbeben in den Rhein- und Niederlanden vom 23.
 Forel F A 1881: Archives Scientifiques Phys. Nat., Vol. 6. Februar 1828. Annalen der Physik und Chemie, Band 13 (89)
 Hamilton W 1783: Philosophical Transactions of the Royal Society, 73, 169–208.
 Jeitteles L H 1859: Bericht über das Erdbeben am 15. Januar 1858 in der Karpathen und Sudeten. In: Sitzungsberichten der Mathem. Naturwiss. Classe der Wissenschaftlichen Akademie in Wien, 35, 511–589.
 Kitaibel P, Tomcsányi A 1814: Dissertatio de terrae motu Morensi anno 1810, Budae
 Mallet R 1862: Great Neapolitan earthquake of 1857: the first principles of observational seismology
 Musson R 1994: A short history of intensity and intensity scales (http://www.earthquakes.bgs.ac.uk/hazard/History_intensity.htm)

- Nöggerath J J 1847: Das Erdbeben vom 29 Juli 1846 im Rheingebiet und der benachbarten Ländern, Bonn
- Ormay S 1875: The earthquake of Jászberény in 1868. Contributions of 2nd department Mathematics and Natural Sciences of the Hungarian Academy of Sciences (in Hungarian). 13, 5–17.
- Réthly A 1960: Report on the earthquake of Mór in the year 1810. Akadémia Kiadó, Budapest
- Sieberg A 1923: Erdbebenkunde, Gustav Fischer, Jena
- Vivenzio G 1786: Istoria de' tremuoti avvenuti nella Provincia della Calabria ulteriore, e nella Città di Messina nell' anno 1783, Naples
- Volger G H O 1856: Untersuchungen über das letztjährige Erdbeben in Central-Europa. Mitteilungen aus Justus Perthes Geographischer Anstalt. H Haessel, Gotha, Germany, 85–102.
- Volger G H O 1857: Untersuchungen über das Phänomen der Erdbeben in der Schweiz. J Perthes, Gotha, Germany
- von Seebach K A L 1973: Das mitteldeutsche Erdbeben vom März 1872. Ein Beitrag zu der Lehre von den Erdbeben, Leipzig